TOPIC NAME : Assignment 1 DAY : _____

TIME : DATE : / /

Answer to the question - 1

4F97 H (+)

(+) 3012 H (+)

7F49 H

0111 ~ 0100 1001

∴ 2F = 0

SF = 0

PF = 0

OF = 0

CF = 0

Answer to the question - 2

(a)

For DS, physical address = DSR * 10 + BX

= 34A9 * 10 + 8722

= 34A90 + 8722

= 3D1B2 H

∴ PA = S.A + offset

⇒ S.A = P.A - offset

⇒ S.A = 3D1B2 - 7B85

⇒ S.A = 35C2D H

Since the starting Address is invalid the segment value is also invalid.

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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

(b) Physical address = 3D1B2 H

∵ segment * 10 + offset = 3D1B2

⇒ segment * 10 = 3D1B2 - offset

⇒ segment * 10 = 3D1B2 - 45D2

⇒ segment * 10 = 38BE0 (valid)

⇒ segment = $\frac{38BE0}{10}$

⇒ segment = 38BE H

(a) Physical address = 3D1B2 H

segment = 7D92 H

∴ P.A = segment * 10 + offset

⇒ offset = P.A - segment * 10

⇒ offset = 3D1B2 - 7D92 * 10

⇒ offset = FFFBF892

This is a negative value and offset cannot be negative. Also offset can also be only 16 bits

∴ offset is invalid

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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Answer to the question - 3

Given, total memory capacity = 128 MB

for RAM [A]

Data Bus = 8 bits

∴ RAM size = $2^{A.B} * D.B$

⇒ 128 MB = $2^{A.B} * 8$ bit

⇒ $128 \times 2^{20} \times 8 = 2^{A.B} * 8$

⇒ $2^{A.B} = 27$ bits

for RAM [B],

Data Bus = 16 bits

∴ 128 MB = $2^{A.B} * 16$ bits

⇒ $128 \times 2^{20} \times 8 = 2^{A.B} \times 16$

⇒ $2^{A.B} = \frac{128 \times 2^{20} \times 8}{16}$

⇒ A.B = 26 bits.

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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Answer to the question no-4

(a) let, Data transfer rate = x kB/s

= x × 1024

$\therefore 133395083\ ns = 150\ ns + \frac{2048 \times 10^9}{x \times 1024}\ ns + 550\ ns$

$\Rightarrow 133394383 = \frac{2048 \times 10^9}{x \times 1024}$

$\Rightarrow x = 149.98$

$\Rightarrow x = 150$ kB/s

(b) Data transfer is happening in this manner,

$= 1+2+3+\ldots+n$

So $1+2+3+\ldots+n = 7875$

$\Rightarrow \frac{n(n+1)}{2} = 7875$

$\Rightarrow n^2+n = 7875 \times 2$

$\Rightarrow n = \frac{-1 \pm \sqrt{(1)^2 - 4 \times 1 \times (-15750)}}{2 \times 1}$

$\Rightarrow n = 125$ and $n = -126$

$\therefore$ time $= \left(150\ ns + \frac{125 \times 10^9}{150 \times 1024} + 550\ ns\right) \times \frac{7875}{125}$

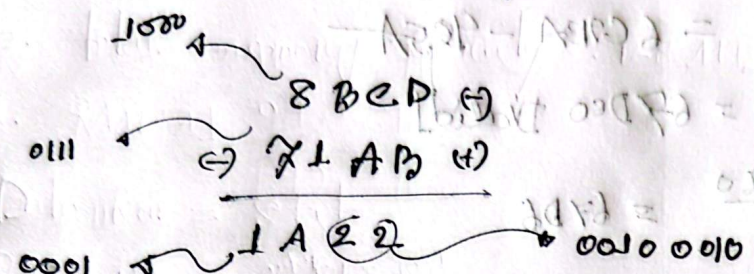
$= 5131363.25\ ns$

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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Answer to the question no-5



CF = 0 [no carry]

ZF = 0 [Result is nonzero]

SF = 0 [The MSB of the result is zero]

PF = 1 [The num of 1's is even]

OF = 1 [(+) - (+) = +]

Answer to the question no-6

(a) Given, 68E2H : 3B9AH

P.A = segment * 10 + offset

⇒ P.A = 68E2 * 10 + 3B9A

⇒ P.A = 6C9BAH

P.A = 6C9BA

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

segment * 10 + offset = 6C9BA

⇒ segment * 10 = 6C9BA - 4E5A

⇒ segment * 10 = 67DC0 [valid]

⇒ segment = $\frac{67DC0}{10}$ = 67DC

(b) PA = 6C9BA H

⇒ segment * 10 + offset = 6C9BA

⇒ segment * 10 = 6C9BA - 2D98

⇒ segment * 10 = 64C22 [invalid]

Since the starting address is invalid therefore segment value is also invalid.

(c) PA = 6C9BA

segment = 26FF

∴ PA = segment * 10 + offset

⇒ offset = PA - segment * 10

⇒ offset = 6C9BA - 26FF * 10

⇒ offset = 499CA

offset is invalid since it has 20 bits.

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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Answer to the question no-7

For segment 1,

S.A = segment * 10

⇒ SA = 8600 * 10

⇒ SA = 86000H

E.A = 86000 + FFFF

⇒ EA = 95FFFH

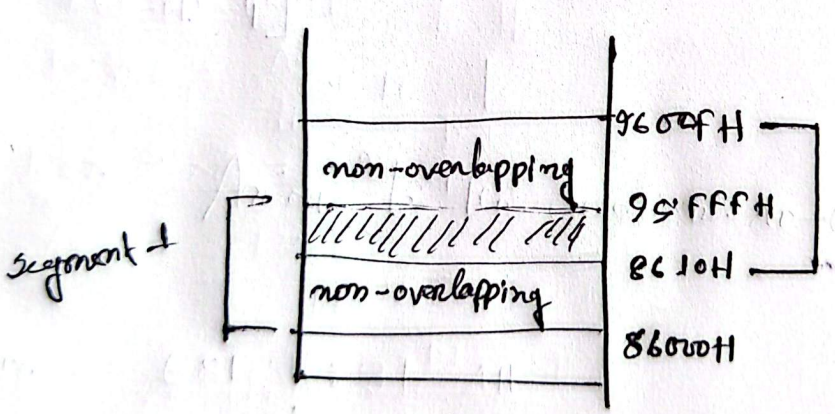
For segment 2,

SA = 8601 * 10

⇒ S.A = 86010H

EA = 86010 + FFFF

⇒ EA = 960FH



∴ The total number of non overlapping physical address dots in both segment is = (8610 - 8600) + (960F - 95FFF)

= 10H + 10H

= 20H Ans.